matching $I(w_1') = I(w_2')$ holds for any consequence $w_1' \simeq w_2'$. In particular, if one has $I(w_1) = I(w_2)$ and $I(w_1') \neq I(w_2')$, then the law $w_1 \simeq w_2$ does not imply the law $w_1' \simeq w_2'$.

A simple example of a matching invariant is the multiplicity $(n_x)_{x \in X}$ of variables of a word: if w_1, w_2 have all variables x appear the same number of times n_x in both words, then any rewriting of a word w using the law $w_1 \simeq w_2$ will preserve this property. Hence, if w_1', w_2' do not have that each variable appear the same number of times in both words, then $w_1 \simeq w_2$ cannot imply $w_1' \simeq w_2'$. For instance, the commutative law E43 cannot imply the left-absorptive law E4.

One source of matching invariants comes from the free magma $\mathcal{M}_{X,\Gamma}$ of a theory:

Proposition 6.1 (Free magmas and matching invariants)**.** *Let $\iota_{X,\Gamma} \colon X \to M_{X,\Gamma}$ be the map associated to the free magma $\mathcal{M}_{X,\Gamma}$ for a theory Γ. Then the map $I \colon M_X \to M_{X,\Gamma}$ defined by $I(w) \coloneqq \varphi_{\iota_{X,\Gamma}}(w)$ is an invariant.*

Proof. Suppose that $w_1 \simeq w_2$ entails $w_1' \simeq w_2'$, and that $I(w_1) = I(w_2)$. For any $f \colon X \to \mathcal{M}_{X,\Gamma}$, the two maps $\varphi_f, \varphi_{f,\Gamma} \circ \varphi_{\iota_{X,\Gamma}} \colon \mathcal{M}_X \to \mathcal{M}_{X,\Gamma}$ are both homomorphisms that extend f, hence agree by the universal property of $\mathcal{M}_X$, as displayed by the following commutative diagram:

$$\begin{tikzcd}
 & X \arrow[dl, hook] \arrow[d, "\iota_{X,\Gamma}"'] \arrow[dr, "f"] & \\
\mathcal{M}_X \arrow[r, "I=\varphi_{\iota_{X,\Gamma}}"] \arrow[rr, bend right, "\varphi_f"'] & \mathcal{M}_{X,\Gamma} \arrow[r, "\varphi_{f,\Gamma}"] & \mathcal{M}_{X,\Gamma}
\end{tikzcd}$$

In particular, the hypothesis $I(w_1) = I(w_2)$ implies that $\varphi_f(w_1) = \varphi_f(w_2)$ for all $f \colon X \to M_{X,\Gamma}$; that is to say, the magma $\mathcal{M}_{X,\Gamma}$ satisfies the law $w_1 \simeq w_2$, and hence also $w_1' \simeq w_2'$ by hypothesis. Thus $\varphi_{\iota_{X,\Gamma}}(w_1') = \varphi_{\iota_{X,\Gamma}}(w_2')$, which gives $I(w_1') = I(w_2')$ as required. □

Example 6.2. If we take $\Gamma = \{\mathrm{E4}\}$ to be the theory of the left-absorptive law E4 as described in Example 2.2, then the matching invariant $I(w)$ produced by Proposition 6.1 is the leftmost letter of the alphabet X appearing in the word; for instance $I((\mathrm{x} \diamond \mathrm{y}) \diamond \mathrm{z}) = \mathrm{x}$. Thus, for example, the left-absorptive law E4 cannot imply the right-absorptive law E5.

Example 6.3. If we take $\Gamma = \{\mathrm{E43}, \mathrm{E4512}\}$ to be the theory of the commutative law E43 and the associative law E4512, then by Example 2.1, the associated invariant $I(w) = \sum_{x \in X} n_x e_x$ is the formal sum of all the generators e_x appearing in the word w, in the free abelian semigroup generated by those generators. This recovers the preceding observation that the multiplicities $(n_x)_{x \in X}$ form a matching invariant.

Example 6.4. Let $n \geq 1$ be a positive integer, and consider the theory $\Gamma = \{\mathrm{E43}, \mathrm{E4512}, \mathrm{E}_n\}$ consisting of the previous theory $\{\mathrm{E43}, \mathrm{E4512}\}$ together with the order-n law $L_x^n y = y$. One can check that the free magma $\mathcal{M}_{X,\Gamma}$ can be described as the free abelian group of exponent n with generators $e_x, x \in X$, with associated map $\iota_{X,\Gamma} \colon x \mapsto e_x$. The associated matching invariant $I(w) = \sum_{x \in X} n_x e_x$ is essentially the multiplicities $(n_x \bmod n)_{x \in X}$ modulo n, which gives a slightly stronger criterion than the preceding matching invariant for refuting implications. For example, the cubic idempotent law $\mathrm{x} \simeq (\mathrm{x} \diamond \mathrm{x}) \diamond \mathrm{x}$ (E23) has matching

invariants $e_x = 3e_x$ in the $n = 2$ case, and hence does not imply the idempotent law $\mathrm{x} \simeq \mathrm{x} \diamond \mathrm{x}$ (E3) since $e_x \neq 2e_x$ in the $n = 2$ case.

In practice, we found that these invariants could be used to establish a significant fraction of the non-implications in the implication graph, although in most cases these non-implications could also be established by other means, for instance through consideration of small finite counterexamples, especially small models of Γ.

Remark 6.5. One can also obtain matching invariants from the free objects associated to theories that involve additional operations beyond the magma operation $\diamond$, such as an identity element or an inverse operation. We leave the precise generalization of Proposition 6.1 to such theories to the interested reader.

6.3. **Canonization.** One possible way of obtaining refutations of a given implication $\mathrm{E} \models \mathrm{E}'$ between equational laws is by building a special kind of syntactic model, via certain involutions on elements of the free magma $\mathcal{M}_X$ we call *canonizers*.

Definition 6.6. A function $C\colon M_X \to M_X$ is a *canonizer* for an equation E if

(1) C is computable.
(2) if $w_1 \sim_{\mathrm{E}} w_2$, then $C(w_1) = C(w_2)$.

In fact, a canonizer is simply a matching invariant with target in M_X.

We describe a concrete strategy for building such Cs, which will require a number of definitions.

Definition 6.7. Let $R\colon M_X \to M_X$ be an arbitrary function.

- We say that R is *weakly collapsing* if for every word w, $R(w)$ is a (not necessarily proper) sub-word of w.
- A function $\theta\colon X \to M_X$ is called a *substitution*, and we can extend θ to an endomorphism $\varphi_\theta\colon \mathcal{M}_X \to \mathcal{M}_X$ in the usual way. We write $\varphi_\theta w$ instead of $\varphi_\theta(w)$ for application of substitutions.
- If E is the equation $l \simeq r$, and l is not a variable, we say that R is a *weak canonizer* if for any substitution θ, $R(\varphi_\theta l) = \varphi_\theta r$.
- Finally we say that R is *non-overlapping* for E if for every word $w \in M_X$ which is a strict sub-word of l that is not in X, and any substitution θ, $R(\varphi_\theta w) = \varphi_\theta w$.

We can then define $C_R\colon M_X \to M_X$ as follows:

$$C_R(\mathrm{x}) \coloneqq \mathrm{x} \tag{9}$$

$$C_R(w \diamond w') \coloneqq R(C_R(w) \diamond C_R(w')) \tag{10}$$

The following lemma is readily proven by induction on the structure of terms.

Lemma 6.8. *If t is a strict sub-word of l, and R is non-overlapping, then $C_R(\varphi_\theta t) = \varphi_{C_R \circ \theta} t$.*

We now have the following theorem.

Theorem 6.9. *Whenever R is weakly collapsing, a weak canonizer, and non-overlapping, then C_R is a canonizer.*

Proof. Assume $w \sim_{\mathrm{E}} w'$. We proceed by induction over the proof of equality.

The only nontrivial case is $w = \varphi_\theta l$ and $w' = \varphi_\theta r$ for some substitution θ. The case $l = r$ is clear, so we may assume that $l \neq r$. Then we have

$$\begin{aligned} C_R(\varphi_\theta l) &= R(\varphi_{C_R \circ \theta} l) \\ &= \varphi_{C_R \circ \theta} r \\ &= C_R(\varphi_\theta r) \end{aligned}$$

where the first and third equalities follow from the Lemma, observing that r is a strict subterm of l (since R is weakly collapsing and a weak canonizer), and the second from weak canonicity. □

We mention an example to show why this is a useful theorem. Take E to be the equation

$$\mathrm{y} \diamond (\mathrm{x} \diamond (\mathrm{y} \diamond (\mathrm{y} \diamond \mathrm{y}))) \simeq \mathrm{x}.$$

We can take R to be the transformation which sends a term of the form $w \diamond (v \diamond (w \diamond (w \diamond w)))$ to v for any two words v, w, and leaves all other words unchanged. It is then somewhat easy to show that this transformation satisfies the conditions of Theorem 6.9, and so we have a canonizer C_R. This can be used, e.g. to refute the implication of $\mathrm{x} \simeq (\mathrm{x} \diamond \mathrm{x}) \diamond (\mathrm{x} \diamond (\mathrm{x} \diamond \mathrm{x}))$ from E.

As a conclusion for this section, we note that a very general strategy for building canonizers comes from the theory of *rewrite systems*, see e.g. Baader & Nipkow [7]. In that setting one defines rewriting as a transformation on words (or terms), and if this transformation is *terminating* and *confluent* (intuitively, rewrites cannot go on forever, or diverge forever), then one may simply pick the transformation which sends a term to its normal form as a canonizer.

Though we note that the non-overlapping criterion seems very similar to the notion of orthogonality in rewriting, we leave the investigation of the precise relationship of the classical theory with the above technique as future work.

6.4. **Unique factorization.** In general, the free magma $\mathcal{M}_{X,\mathrm{E}}$ for a given equational law E, which we can canonically define as $\mathcal{M}_X / \sim_{\mathrm{E}}$, is hard to describe explicitly; indeed, from the undecidability of implications between equational laws, such a magma cannot be computably described for arbitrary E. Nevertheless, for some laws it is possible to obtain some partial understanding of $\mathcal{M}_{X,\mathrm{E}}$ from a syntactic perspective. For instance, if we can refute the equivalence $w_1' \sim_{\mathrm{E}} w_2'$ by constructing a counterexample magma $\mathcal{M}$ that satisfies E but not $w_1' \simeq w_2'$, then this implies that the representatives $\iota_{X,\mathrm{E}}(w_1'), \iota_{X,\mathrm{E}}(w_2')$ of w_1', w_2' in $\mathcal{M}_{X,\mathrm{E}}$ are distinct.

We illustrate this approach with equations E of the left-absorptive form

$$\mathrm{x} \simeq \mathrm{x} \diamond f(\mathrm{x}, \mathrm{y}, \mathrm{z}) \tag{11}$$